# PYTHON PROGRAMMING — CLASS TEST

Name: _____ Date: _____

Instructions: Answer all questions. Write code clearly where requested. Each topic contains five questions.

Total: 80 questions across 16 topics | Files and Django excluded

## Section 1: The Way of the Program

1. What is a computer program?

2. Explain the difference between source code and output.

3. Name two types of errors that can occur in a Python program.

4. What is the role of the Python interpreter?

5. Write a Python statement that displays Welcome to Python.

## Section 2: Variables, Expressions and Statements

1. What is a variable? Give one valid Python variable name.

2. State the output of x = 4; x = x + 3; print(x).

3. Explain the difference between an expression and a statement.

4. Write an expression to calculate the area of a rectangle using length and width.

5. What is wrong with the variable name 2score?

## Section 3: Functions

1. What is a function in Python?

2. Write a function named greet that prints Hello.

3. Explain the purpose of parameters in a function.

4. What is the difference between defining and calling a function?

5. Write a function call that passes 5 to a function named double.

## Section 4: Conditionals

1. What is the purpose of an if statement?

2. Write an if statement that prints Adult when age is at least 18.

3. Explain the difference between elif and else.

4. What does the expression 7 > 3 and 2 > 5 evaluate to?

5. Write a conditional that prints Even if a number is divisible by 2 and Odd otherwise.

## Section 5: Recursion

1. What is recursion?

2. What is a base case in a recursive function?

3. Explain what may happen if a recursive function has no reachable base case.

4. Write a recursive function that returns the factorial of a non-negative integer.

5. State one difference between recursion and iteration.

## Section 6: Fruitful Functions

1. What is a fruitful function?

2. How does return differ from print inside a function?

3. Write a function square(n) that returns n multiplied by itself.

4. What value is assigned to result after result = max(3, 9)?

5. Write a function that accepts two numbers and returns their average.

## Section 7: Iteration

1. What is iteration in programming?

2. Write a for loop that prints numbers 1 through 5.

3. Explain one difference between for and while loops.

4. What is the purpose of break in a loop?

5. Write a while loop that counts down from 3 to 1.

## Section 8: Strings

1. What is a string in Python?

2. What is the output of print("Python"[0])?

3. Explain what string slicing means.

4. Write Python code that converts "hello" to uppercase.

5. What is the result of len("coding")?

## Section 9: Lists

1. What is a Python list?

2. Write a list containing three programming languages.

3. How do you access the second element of a list named items?

4. Explain the difference between append() and remove().

5. What will print([10, 20, 30][-1]) display?

## Section 10: Dictionaries

1. What is a Python dictionary?

2. Create a dictionary with keys name and age.

3. How do you retrieve the value stored under the key name?

4. What happens when an existing dictionary key is assigned a new value?

5. State one difference between a dictionary and a list.

## Section 11: Tuples

1. What is a tuple in Python?

2. Write a tuple containing the numbers 2, 4 and 6.

3. State one key difference between a tuple and a list.

4. What is the output of print((5, 8, 11)[1])?

5. Can you directly change an element of a tuple? Explain.

## Section 12: Classes and Objects

1. What is a class in Python?

2. What is an object?

3. Write a simple class named Student with no attributes or methods.

4. Explain the relationship between a class and an instance.

5. What is the purpose of the __init__ method?

## Section 13: Classes and Functions

1. How can a function be used to create an object of a class?

2. Write a function make_student() that returns an instance of a Student class.

3. What is the difference between a function defined outside a class and one defined inside it?

4. Explain why a function might accept an object as an argument.

5. Write a function that accepts an object with a name attribute and prints its name.

## Section 14: Classes and Methods

1. What is a method in Python?

2. What does self usually refer to in an instance method?

3. Write a class Dog with a method bark() that prints Woof.

4. How do you call the bark method on an object named pet?

5. Explain the difference between an instance attribute and a method.

## Section 15: Inheritance

1. What is inheritance in object-oriented programming?

2. Write a class Car that inherits from a class Vehicle.

3. State one benefit of inheritance.

4. What is method overriding?

5. Explain the purpose of super() in a subclass.

## Section 16: Analysis of Algorithms

1. What does algorithm efficiency mean?

2. What is time complexity?

3. Explain the meaning of O(n) in simple terms.

4. Which usually scales better for large n: O(n) or $O(n^2)$? Explain.

5. State one reason why space complexity matters.